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II/IV-B.Tech(EVEN Sem), Academic Year: 2025-2026
B.Tech(CSE, CSIT &ME), 2024 Batch II/IV, EVEN Semester
Subject Code: 24MT2019 / TITLE: Probability and Statistics
SESSIONWISE PROBLEMS (CO-2)

SESSION-7 :

(Pearson's correlation coefficient & Spearman's Rank Correlation Method)

1. The following data reflects fertilizer used (in tons.) and corresponding productivity of the crop (in tons.). Obtain correlation coefficient by applying Karl Pearson's formula to the following data:

Fertilizer Used (Tonnes)	15	18	20	24	30	35	40	50
Productivity (Tonnes)	85	93	95	105	120	130	150	160

2. Obtain a significant correlation coefficient between the heights and weights of individuals given below:

Height in Inches	57	59	62	63	64	65	55	58	57
Weight in Lbs	113	117	126	126	130	129	111	116	112

3. The ranks of the 15 students in two subjects A and B are given below, the two numbers within the brackets denoting the ranks of the same student in A and B respectively.
(1, 10), (2, 7), (3, 2), (4, 6), (5, 4), (6, 8), (7, 3), (8, 1), (9, 11), (10, 15), (11, 9), (12, 5), (13, 14), (14, 12), (15, 13).
Apply Spearman's Rank correlation formula to find the rank correlation coefficient.

4. A sample of 12 fathers and their elder sons gave the following data about their elder sons. Obtain the coefficient of rank correlation. Also draw a scatter diagram for this data:

Fathers	65	63	67	64	68	62	70	66	68	67	69	71
Sons	68	66	68	65	69	66	68	65	71	67	68	70

SESSION-8: (Linear Regression: Regression line 'y' on 'x' and 'x' on 'y')

1. Obtain the most likely production corresponding to a rainfall 40 from the following data by fitting a regression line:

	Rain fall (X)	Production (Y)
Average	30	500 Kgs
Standard Deviation	5	100 Kgs
Coefficient of Correlation	0.8	

2. From the following bi-variate data (Which reflects the data regarding the Hospitals(x) and number of Doctors(y) available in a city):

X	1	5	3	2	1	1	7	3
Y	6	1	0	0	1	2	1	5

a) Obtain a Regression line of Y on X and hence predict, Y if X = 10.
b) Obtain a Regression line of X on Y and hence predict, X if Y= 2.5.
c) Calculate the Correlation Coefficient.

3. The following information came from Security Officer (Number of Security personnel attended their duties(x) Vs Number of Security personnel who are on leave(y)) to estimate the i) Correlation Coefficient ii) Standard deviation of y (σ_y)

$b_{xy} = 0.85,\ b_{yx} = 0.89,\ \sigma_x = 3.$

4. Find the most likely price in Delhi corresponding to the price of Rs.70 at Chennai from the following data:

	Chennai	Delhi
Average Price	65	67
Standard Deviation	2.5	3.5

Correlation coefficient between the prices of commodities in the two cities is 0.8.

SESSION-9: (Regression Line and Second Degree Polynomial by the method of Least Squares)

1. An article in the *Tappi Journal* presented data on green liquor Na2S concentration (in grams per liter) and paper machine production (in tons per day). The data (read from a graph) follow:

Y	40	42	49	46	44	48	46	43
X	825	830	890	895	890	910	915	960

(a) Fit a simple linear regression model with y = green liquor Na2S concentration and x = production, by least squares model.
(b) Find the fitted value of y corresponding to x = 912.
(c) Find the mean green liquor Na2S concentration when the production rate is 950 tons per day.

2. An article in the *Journal of Sound and Vibration* (1991) described a study investigating the relationship between noise exposure and hypertension. The following data are representative of those reported in the article.

y	1	0	1	2	5	1	4	6	7	6
x	60	63	65	70	70	70	80	90	100	100

(a) Draw a scatter diagram of '*y*' (blood pressure rise in milli meters of mercury) versus '*x*' (sound pressure level in decibels). Does a simple linear regression model seem reasonable in this situation?
(b) Fit the simple linear regression model using least squares.
(c) Find the predicted mean rise in blood pressure level associated with a sound pressure level of 85 decibels.

3. Regression methods were used to analyze the data from a study investigating the relationship between roadway surface temperature (x) and pavement deflection (y). Summary quantities were $n = 20$, $\Sigma y = 12.75$, $\sum y^2 = 8.86$, $\Sigma x = 1478$, $\sum x^2 = 143{,}215.8$, and $\Sigma x y = 1083.67$.

(a) Calculate the least squares estimates of the slope and intercept.
(b) Use the equation of the fitted line to predict what pavement deflection would be observed when the surface temperature is 85°F.
(c) What is the mean pavement deflection when the surface temperature is 90°F?
(d) What change in mean pavement deflection would be expected for a 1°F change in Surface temperature?

4. An article in the *Journal of Environmental Engineering*, reported the results of a study on the occurrence of sodium and chloride in surface streams in central Rhode Island. The following data are chloride concentration '*y*' (in milligrams per liter) and roadway area in the watershed '*x*' (in percentage).

x	0.19	0.15	0.57	0.70	0.67	0.63	0.47	0.7	0.6	0.78
y	4.4	6.6	9.7	10.6	10.8	10.9	11.8	12.1	14.3	14.7

(a) Draw a scatter diagram of the data.
(b) Fit the simple linear regression model, $y = a + b x + c x^2$ using the method of least squares.
(c) Estimate the mean chloride concentration for a watershed that has 1% roadway area.
(d) Find the fitted value corresponding to $x = 0.47$.

5. In an article in *Statistics and Computing*, Carlin and Gelfand investigated the age(x) and length (y) of 10 captured dugongs (sea cows).

x	1.0	1.5	2.5	4.0	7.0	8.0	9.0	9.5	29.0	31.5
y	1.8	1.85	2.02	2.27	2.15	2.26	2.47	2.56	2.72	2.57

(a) Fit a second degree polynomial to the above data by applying method of least squares.
(b) Estimate the mean length of dugongs at age 11.

SESSION-10: (Non-linear regression: Fitting Exponential and Power curve)

1. A simply supported beam carries a concentrated load 'x' (lbs) at its mid-point corresponding to various values of 'x', the maximum deflection 'y' (inches) is measured. The data are given below. Obtain a law of the type, $y = a e^{bx}$.

x	2	4	6	8	10
y	4.077	11.084	30.128	81.897	222.62

2. A chemical company, wishing to study the effect of extraction time on the efficiency of an extraction operation, obtained the data shown in the following table. Fit a curve of the form $y = a b^x$ by the method of least squares.

x	0	1	2	3	4	5
y	10	21	35	59	92	200

3. The following are measurements of air velocity and evaporation coefficient of burning fuel droplets in an impulse engine. Fit an exponential curve to the following data

Air velocity (cm/sec) (x)	20	60	100	140	180	220	260
Evaporation coefficient (mm^2/sec) (y)	0.18	0.37	0.35	0.78	0.56	0.75	1.18

4. A van rental company records the number of kilometers driven (x) and the amount charged(y) for different customers.

Distance (x in km)	10	20	30	40	50
Charge (y in ₹)	250	300	450	500	650

Determine the equation of a power curve of the form, $y = a b^x$ representing the charge based on distance and also obtain the charge when distance is 200km.

SESSION-11: (BASIC CONCEPTS OF PROBABILITY)

1. Out of 15 items 4 are not in good condition. Four items are selected at random. Find the Probability that i) all are good ii) two are not good.

2. The supervisor of a group of 20 construction workers wants to get the opinion of 2 of them (to be selected at random) about certain new safety regulations. If 12 workers favour the new regulations and the other 8 are against them, what is the probability that both of the workers chosen by the supervisor will be against the new safety regulations?

3. A B.Tech graduate applies for a job in two firms X and Y. The probability of his being selected in firm X is 0.7 and being rejected at Y is 0.5. The probability of at least one of his applications being rejected is 0.6. What is the probability that he will be selected in one of the firms?

4. Three newspapers A, B and C are published in a certain city. It is estimated from a survey that of the adult population: 20% read A, 16% read B, 14% read C, 8% read both A and B, 5% read both A and C, 4% read both B and C, 2% read all three. Find what percentage read at least one of the papers?

5. A letter of the English alphabet is chosen at random. Calculate the probability that the letter so chosen i) is a vowel ii) precedes 'm' and is a vowel iii) follows 'm' and is a vowel.

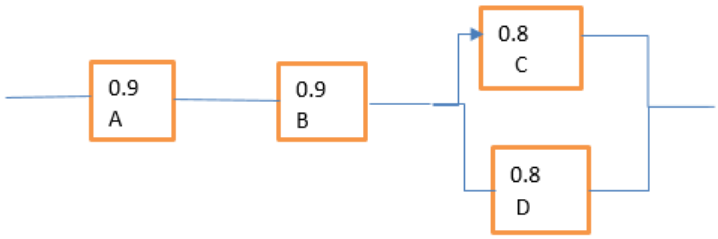
SESSION-12: (CONDITIONAL PROBABILITY AND INDEPENDENT EVENTS)

1. From a city population, the probability of selecting i) a male or a smoker is 7/10, ii) a male smoker is 2/5 and iii) a male, if a smoker is already selected is 2/3. Find the probability of selecting (a) a non-smoker (b) a male and (c) a smoker, if a male is first selected.

2. One shot is fired from each of the three guns E_1 , E_2 , E_3 denote the events that the target is hit by the first, second and third guns respectively. If $P(E_1) = 0.5$, $P(E_2) = 0.6$ and $P(E_3) = 0.8$ and E_1 , E_2 , E_3 are independent events, find the probability that i) exactly one hit is registered, and ii) at least two hits registered.

3. A manager has two assistants and he bases his decision on information supplied independently by each one of them. The probability that he makes a mistake in his thinking is 0.005. The probability that an assistant gives wrong information is 0.3. Assuming that the mistakes made by the manager are independent of the information given by the assistants.

4. An electrical system consists of four components as illustrated in the following figure. The system works if components A and B and either of the components C or D works. The reliability (probability of working) of each component is also shown in the figure. Obtain the probability that a) The entire system works and b) the component C doesn't work, given that the entire system works. Assume that the four components work independently.



SESSION-13: (Total Probability and Bayes' Formula)

1. A business man goes to hotels X, Y, Z, 20%, 50%, 30% of the time respectively. It is known that 5%, 4%, 8% of the rooms in X, Y, Z hotels have faulty plumbing. What is the probability that the business man's room having faulty plumbing is assigned to hotel 'Z'?

2. The chances that doctor A will diagnose a disease X correctly is 60%. The chances that a patient will die by his treatment after correct diagnosis is 40% and the chance of death by wrong diagnosis is 70%. A patient of doctor A, who had disease X, died. What is the chance that his disease was diagnosed correctly?

3. From a vessel containing 3 white and 5 black balls, 4 balls are transferred into an empty vessel. From this vessel a ball is drawn and is found to be white. What is the probability that out of four balls transferred 3 are white and 1 is black?

4. In a bolt factory machines A, B, C manufacture 20%, 30%, and 50% of the total of their output and 6%, 3%, 2% are defective. A bolt is drawn at random and found to be defective. Obtain the probabilities that it is manufactured from a) Machine A b) Machine B and c) Machine C.